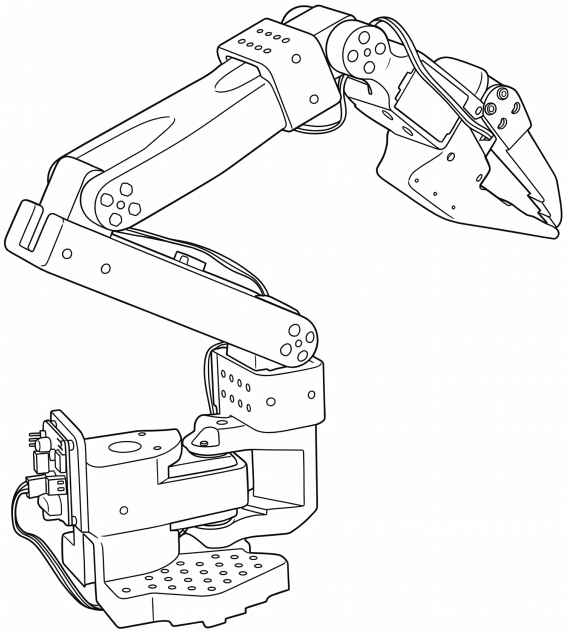
Introduction to Robotic Manipulation Project



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Phase 1 – Kinematics Modelling

1.1 Introduction

The goal of the project is a Cartesian control system for the LeRobot SO-101 arm based on inverse kinematics, developed and validated in MuJoCo before deployment on the physical robot. **Phase 1** covers the kinematic model, starting from the geometry of the arm, coordinate frame, DH description and a Product of Exponentials (PoE) forward kinematics (FK), validated against the simulation and the physical robot. The SO-101 has six servomotors, but only five of them move the end-effector, which are `shoulder_pan`, `shoulder_lift`, `elbow_flex`, `wrist_flex`, `wrist_roll`. The sixth only opens and closes the moving jaw and does not change the pose of the TCP, so it is left out of the kinematic chain. All geometric data are taken from the MuJoCo model `so101_new_calib.xml`, which is built from the CAD components, and roughly checked on the physical robot with a ruler. Joint angles q_i are the MuJoCo joint values in radians: $\mathbf{q} = \mathbf{0}$ is the home pose of **Figure 1.1**, with the upper arm almost vertical and the forearm almost horizontal. Lengths are given in millimeters.

1.2 SO-101: Arm Configuration

Table 1.1 lists the position and direction of every joint axis in the world frame at $\mathbf{q} = \mathbf{0}$, printed by option 8 (Extract PoE) of `run.sh`. The pan axis points along $-z$ and the roll axis along $-x$, so a positive q_1 or q_5 turns the arm clockwise about the corresponding DH axis. This is the main reason why both appear with a minus sign in **Table 1.2**.

Joint	**Position [mm]**			**Axis direction**		
	X	Y	Z	u_x	u_y	u_z
`shoulder_pan`	38.84	0.00	62.40	0	0	−1
`shoulder_lift`	69.23	−18.28	116.60	0	1	0
`elbow_flex`	97.23	−18.28	229.17	0	1	0
`wrist_flex`	232.13	−18.28	234.37	0	1	0
`wrist_roll`	293.23	−0.18	234.37	−1	0	0
`TCP` (`gripperframe`)	391.36	−0.01	226.47		—	

Table 1.1: Joints axes and TCP in world frame at $\mathbf{q} = \mathbf{0}$ from MuJoCo model

The DH lengths follow from the common normals between consecutive axes:

$$d_1 = 116.6 \text{ mm (height of the lift axis)}, \qquad a_1 = 69.23 - 38.84 = 30.4 \text{ mm},$$
$$a_2 = \sqrt{28.0^2 + 112.57^2} = 116.0 \text{ mm}, \qquad a_3 = \sqrt{134.9^2 + 5.2^2} = 135.0 \text{ mm},$$
$$d_5 = 391.36 - 232.13 = 159.2 \text{ mm}, \qquad a_5 = 234.37 - 226.47 = 7.9 \text{ mm}$$

The `wrist-flex` and `wrist-roll` axes intersect, so $a_4 = 0$. Neither link is straight: the `elbow-flex` sits 28.0 mm ahead of the shoulder and the `wrist-flex` 5.2 mm above the elbow, which produces the constant offsets in θ_2, θ_3 and θ_4 (13.97° and 2.21° away from 90° and 0°). The three pitch axes lie in a plane 18.28 mm behind `shoulder-pan` axis, while the roll axis is almost centered ($y = -0.18$ mm); d_2 and d_4 encode this lateral offset. The base frame $\{0\}$ lies on the `shoulder-pan` axis at the bottom of the base, 38.84 mm in front of the MuJoCo world origin and the TCP is 7.9 mm below the roll axis because `gripperframe` is placed on the jaw centerline.

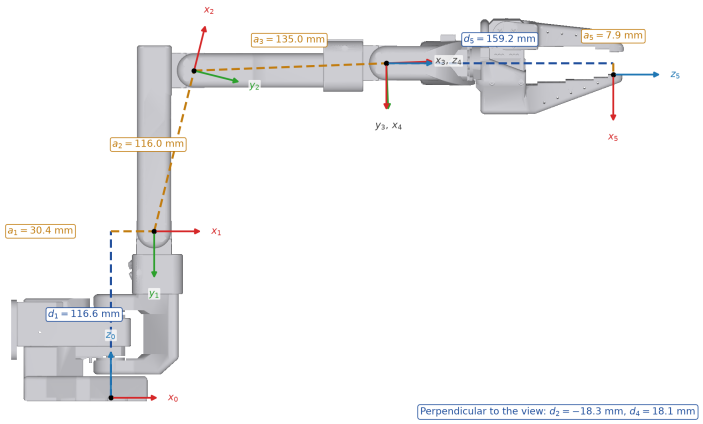


Figure 1.1: DH frames of the SO-101 at $\mathbf{q} = \mathbf{0}$.

Table 1.2: DH parameters of the SO-101 (lengths in mm).

i	θ_i	d_i	a_i	α_i
1	$-q_1$	116.6	30.4	$-\pi/2$
2	$q_2 - 76.03°$	-18.28	116.0	0
3	$q_3 + 73.83°$	0	135.0	0
4	$q_4 + 92.21°$	18.10	0	$\pi/2$
5	$-q_5 + 1.21°$	159.2	7.9	0

θ_i: rotation about z_{i-1}, from x_{i-1} to x_i
d_i: translation along z_{i-1}
a_i: translation along x_i
α_i: rotation about x_i, from z_{i-1} to z_i

1.3 Forward Kinematics (FK)

The FK is implemented with the Product of Exponentials, which uses the joint axes of **Table 1.1** directly. Each joint i has the screw axis $\mathcal{S}_i = (\hat{\omega}_i,\, v_i)$ with $v_i = -\hat{\omega}_i \times p_i$, where $\hat{\omega}_i$ and p_i are the axis direction and a point on the axis at $\mathbf{q} = \mathbf{0}$. The TCP pose is

$$T(\mathbf{q}) = e^{[\mathcal{S}_1]q_1}\, e^{[\mathcal{S}_2]q_2}\, e^{[\mathcal{S}_3]q_3}\, e^{[\mathcal{S}_4]q_4}\, e^{[\mathcal{S}_5]q_5}\, M, \qquad e^{[\mathcal{S}]\theta} = \begin{bmatrix} e^{[\hat{\omega}]\theta} & G(\theta)\, v \\ 0 & 1 \end{bmatrix}$$

with $e^{[\hat{\omega}]\theta} = I + \sin\theta\,[\hat{\omega}] + (1-\cos\theta)[\hat{\omega}]^2$ (Rodrigues) and $G(\theta) = I\theta + (1-\cos\theta)[\hat{\omega}] + (\theta - \sin\theta)[\hat{\omega}]^2$. M is the home pose of the `gripperframe` site: position (391.36, -0.01, 226.47) mm and a 2.79° tilt about x inherited from the CAD. The function `SO_101_fk(q)` in `src/forward_kinematics.py` takes $\mathbf{q} \in \mathbb{R}^5$ in radians and returns $T(\mathbf{q})$ in meters. The constants are in `src/utils/constants.py` and the ones that depend on the configuration are copied and pasted with option 8 of `run.sh`. The DH model of **Table 1.2** describes the same chain as $T(\mathbf{q}) = \mathrm{Trans}_x(38.8\,\mathrm{mm})\, {}^0A_1 \cdots {}^4A_5\, T_{\mathrm{tool}}$, with ${}^{i-1}A_i = \mathrm{Rot}_z(\theta_i)\,\mathrm{Trans}_z(d_i)\,\mathrm{Trans}_x(a_i)\,\mathrm{Rot}_x(\alpha_i)$. The first factor moves the MuJoCo world origin onto the `shoulder-pan` axis and $T_{\mathrm{tool}} \approx \mathrm{Rot}_y(-90°)$ only aligns the axes of frame $\{5\}$ with those of M. We use DH to document the frames and PoE in the code and check in **Section 1.4** that both give the same pose. Since only five joints act on the TCP, the SO-101 cannot reach arbitrary 6-D poses: once the TCP position is fixed, only two orientation components remain free, the pitch of the tool ($q_2 + q_3 + q_4$) and its roll (q_5), while its heading is set by q_1. This shapes **Phase 2** where the inverse kinematics will target the position plus at most a partial orientation.

1.4 Results

Simulation. `SO_101_fk` was compared with the `gripperframe` pose computed by MuJoCo for 2000 random configurations within the joint limits and for 7 named poses (`simulation/fk_validation.py`, `tests/test_fk.py`, all tests passing). The errors are at the level of MuJoCo's output (**Table 1.3**), so the PoE model reproduces the CAD geometry exactly. The DH table matches MuJoCo almost perfectly, the residual coming only from rounding its angle offsets. As a visual check, the DH frames were added to the MuJoCo model as sites and the DH lengths drawn between them in **Figure 1.2**: every z_i stays on its joint axis and the lengths remain constant while the joints move.

Physical Robot. On our arm, the 7 named poses were commanded and held (`hardware/fk_hardware.py`). All the numbers following can be found in `results/data/*.csv`. Wrist and pan joints settle within 0.24° of the command, while the gravity-loaded `shoulder_lift` and `elbow_flex` stay up to 1.45° and 1.32° away. Through the FK, this moves the TCP by a mean of 8.6 mm from the commanded position. A 32 s trajectory through the same poses was also run in MuJoCo and on the robot (`hardware/sim_vs_real.py`, **Figure 1.3**): the real joints follow the commands with RMS errors between 0.3° and 0.8° for pan and wrist and about 1.8° for `shoulder_lift` and `elbow_flex`, which lag behind during motion.

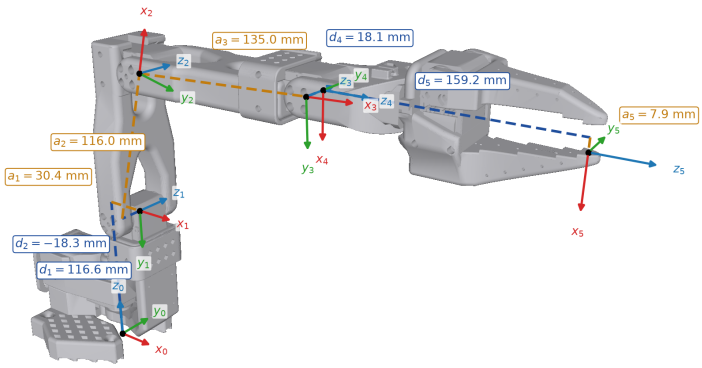


Figure 1.2: DH frames and lengths in MuJoCo

Table 1.3: Validation summary

	Mean	Max
PoE vs MuJoCo, pos. [μm]	0.07	0.08
PoE vs MuJoCo, rot. [μrad]	0.03	0.07
DH vs MuJoCo, pos. [mm]	0.05	0.06
Real, static joint err. [°]	0.40	1.45
Real, static TCP dev. [mm]	8.6	14.8
Real vs cmd., trajectory TCP [mm]	9.1	27.0

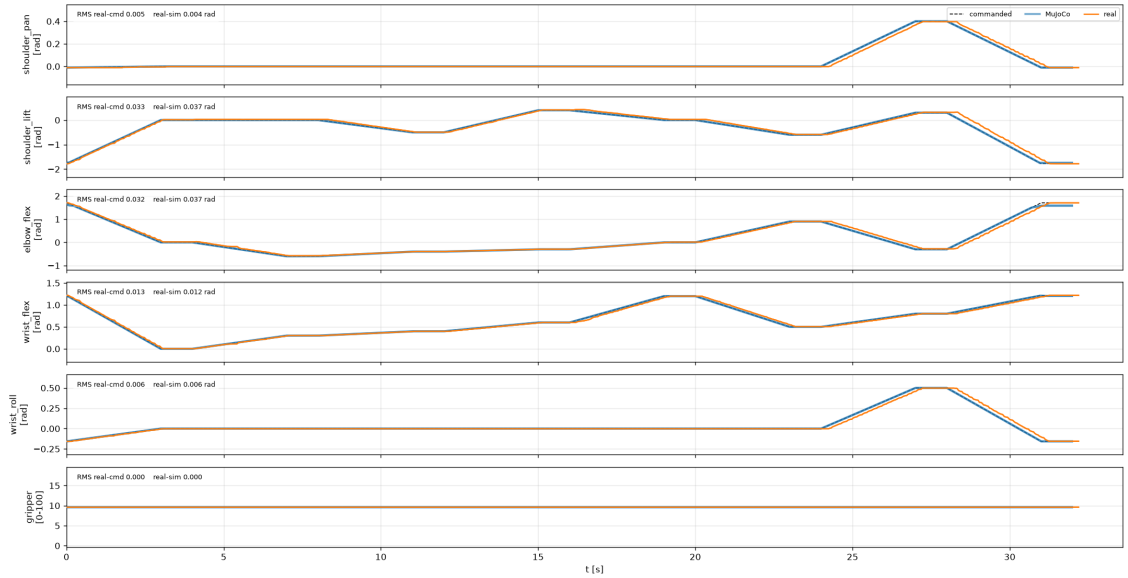


Figure 1.3: Commanded, simulated and real joint angles on the test trajectory.

Limitations. The model describes the nominal CAD geometry. The TCP position of the real arm was not measured independently, so the hardware results validate the joint readings and their effect through the FK, not the link lengths, which were roughly checked with a ruler. The main error source is the servomotors: with the default gains ($P = 32$ and $I = 0$ in `utils/so101_utils.py`), `shoulder-lift` and `elbow-flex` keep a steady-state error under gravity, which **Phase 3** will need to compensate (control phase). The joint limits are those of the MuJoCo model and may be wider than the calibrated ranges and different tool (like a marker) requires a new M.